

Next Greater Element I (LC 496) * *****

LEARN THIS LEARN THIS ***** IMPORTANT*****

```
int[] nextGreaterElements = new int[nums2.length];
for (int i = nums2.length - 1; i >= 0 ; i--) {
    int element = nums2[i];

    // If empty, no next greater element
    if (helperStack.isEmpty()) {
        helperStack.push(element);
        nextGreaterElements[i] = -1;
        continue;
    }

    // If top of stack is greater, it is next greatest
    if (helperStack.peek() > element) {
        helperStack.push(element);
        nextGreaterElements[i] = helperStack.peek();
        continue;
    }

    // Remove all elements smaller than element
    while(helperStack.peek() <= element && !helperStack.isEmpty()) {
        helperStack.pop();
    }

    if (helperStack.isEmpty())
        nextGreaterElements[i] = -1;
    else
        nextGreaterElements[i] = helperStack.peek();

    helperStack.push(element);
}
```

DRY-RUN

nums2 = [1, 3, 4, 1, 2]

$O(n)$

		-1	2	-1

						4

Study Algorithms
...some simple algorithms to help you

it was next greater element to the right of current element

learn how we created the nge in this , it was nge to the right , so we started from right then to left

LEARN LEARN THIS NGE ONE , rest you can come up with

if empty push and mark nge i = -1

if element < top mark nge i = top and push element

while element >= top , you pop

if empty , nge = -1

else nge i = top

push element

503. Next Greater Element II

Solved

Medium Topics Companies

Given a circular integer array `nums` (i.e., the next element of `nums[nums.length - 1]` is `nums[0]`), return *the next greater number for every element in `nums`*.

The **next greater number** of a number `x` is the first greater number to its traversing-order next in the array, which means you could search circularly to find its next greater number. If it doesn't exist, return `-1` for this number.

HINT : you just do it twice on the nums

739. Daily Temperatures

Solved ✓

Medium

Topics

Companies

Hint

Given an array of integers `temperatures` represents the daily temperatures, return *an array* `answer` *such that* `answer[i]` *is the number of days you have to wait after the* i^{th} *day to get a warmer temperature.* If there is no future day for which this is possible, keep `answer[i] == 0` instead.

HINT : push indices to stack , nge = st.top() - i

*****\

1475. Final Prices With a Special Discount in a Shop

Solved ✓

Easy

Topics

Companies

Hint

You are given an integer array `prices` where `prices[i]` is the price of the i^{th} item in a shop.

There is a special discount for items in the shop. If you buy the i^{th} item, then you will receive a discount equivalent to `prices[j]` where `j` is the minimum index such that `j > i` and `prices[j] <= prices[i]`. Otherwise, you will not receive any discount at all.

Return an integer array `answer` where `answer[i]` is the final price you will pay for the i^{th} item of the shop, considering the special discount.

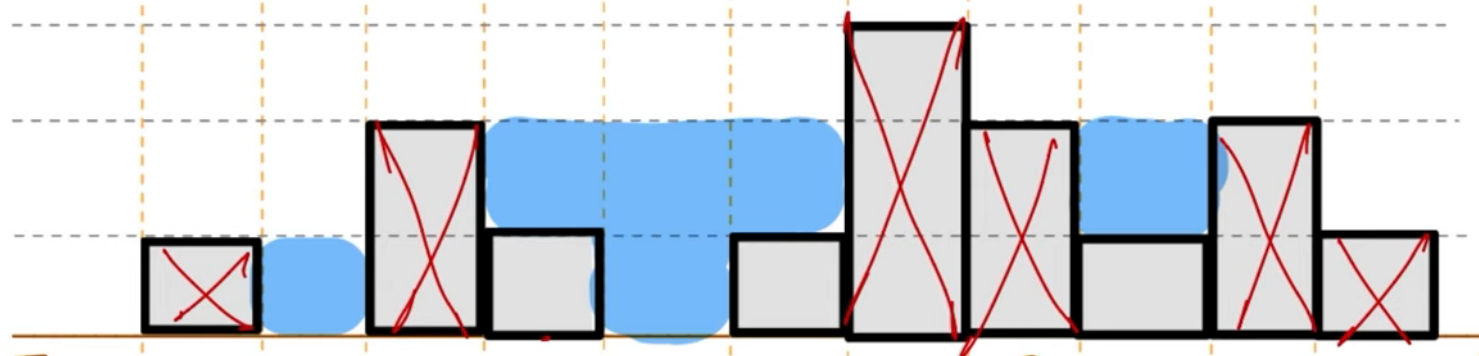
ye bhi same hi , NEXT SMALLER ELEMENT implement karna hai , NGE likho and signn ulta kardo

TRAPPING RAIN WATER *

this solution doesnt uses a stack , we compute both left and right then find the answer , -> 2 scans

$$\text{water}[i] = \min(\text{max_left}, \text{max_right}) - \text{height}[i]$$

[0 , 1 , 0 , 2 , 1 , 0 , 1 , 3 , 2 , 1 , 2 , 1]



left = [0 , 0 , 1 , 1 , 2 , 2 , 2 , 2 , 3 , 3 , 3 , 3]

right = [3 , 3 , 3 , 3 , 3 , 3 , 3 , 2 , 2 , 2 , 1 , 0]

110:

```
int trap(vector<int>& height) {
    int n = height.size();

    vector<int> left(n, 0) , right(n, 0);
    int ans = 0 , water ;
    for(int i=1;i<n;i++){
        left[i] = max(height[i-1] , left[i-1]);
    }

    for(int i=n-2;i>=0;i--){
        right[i] = max(height[i+1], right[i+1]);

        water = min(left[i], right[i]) - height[i];
        if(water>0) ans+=water;
    }

    return ans;
}
```